Lab 3 - Routing Protocol

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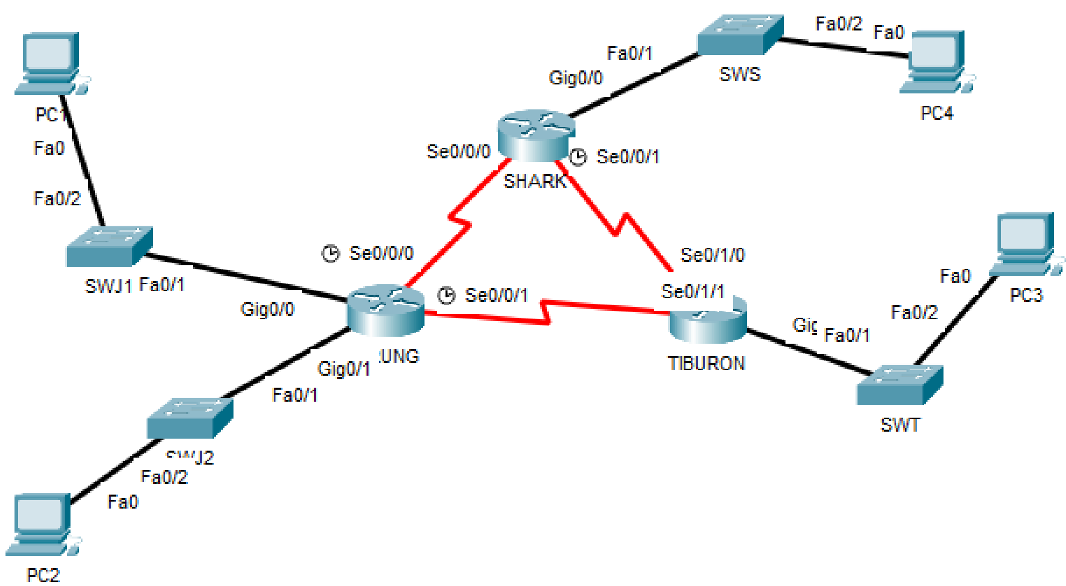
SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

	Usable Host	Subnet Size	Subnet Mask	Network Address	Subnet Range
JERUNG LAN1	20	$2^n \geq 20+2$ $2^5 \geq 22$	$/32-5$ $=/27$	172.18.110.0/27	172.18.110.0 - 172.18.110.31
JERUNG LAN2	30	$2^n \geq 30+2$ $2^5 \geq 32$	$/32-5$ $=/27$	172.18.110.32/27	172.18.110.32 - 172.18.110.63
SHARK LAN	60	$2^n \geq 60+2$ $2^6 \geq 62$	$/32-6$ $=/26$	172.18.110.64/26	172.18.110.64 - 172.18.110.127
TIBURAN LAN	50	$2^n \geq 50+2$ $2^6 \geq 52$	$/32-6$ $=/26$	172.18.110.128/26	172.18.110.128 - 172.18.110.191
JERUNG-SHARK	2	$2^n \geq 2+2$ $2^2 \geq 4$	$/32-2$ $=/30$	172.18.110.192/30	172.18.110.192 - 172.18.110.195
JERUNG-TIBURON	2	$2^n \geq 2+2$ $2^2 \geq 4$	$/32-2$ $=/30$	172.18.110.196/30	172.18.110.196 - 172.18.110.199
SHARK-TIBURON	2	$2^n \geq 2+2$ $2^2 \geq 4$	$/32-2$ $=/30$	172.18.110.200/30	172.18.110.200 - 172.18.110.203

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
    a. CLI
        i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

    b. PT tool
        i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

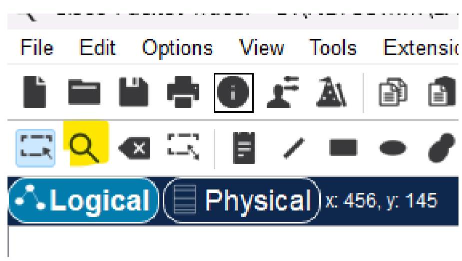


Figure B

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

```
JERUNG>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1
```

```
SHARK>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

```
TIBURON>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/0
L       172.18.110.198/32 is directly connected, Serial0/1/0
C       172.18.110.200/30 is directly connected, Serial0/1/1
L       172.18.110.202/32 is directly connected, Serial0/1/1
```

2. Try to ping PC2 from PC1, paste the results here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=5ms TTL=128
Reply from 172.18.110.30: bytes=32 time=5ms TTL=128
Reply from 172.18.110.30: bytes=32 time<1ms TTL=128
Reply from 172.18.110.30: bytes=32 time<1ms TTL=128

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 2ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time=5ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.
   PC1 can ping PC2 but cannot ping PC4. This is because JERUNG is connected to JERUNG LAN1 where PC1 resides and JERUNG LAN2 where PC2 resides but not the SHARK LAN where PC4 resides. The packet from PC1 can reach the JERUNG router but cannot reach the SHARK router.

5. What needs to be done to ensure all PCs can ping each other successfully?
   Configure the routers with static routes or by a routing protocol.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

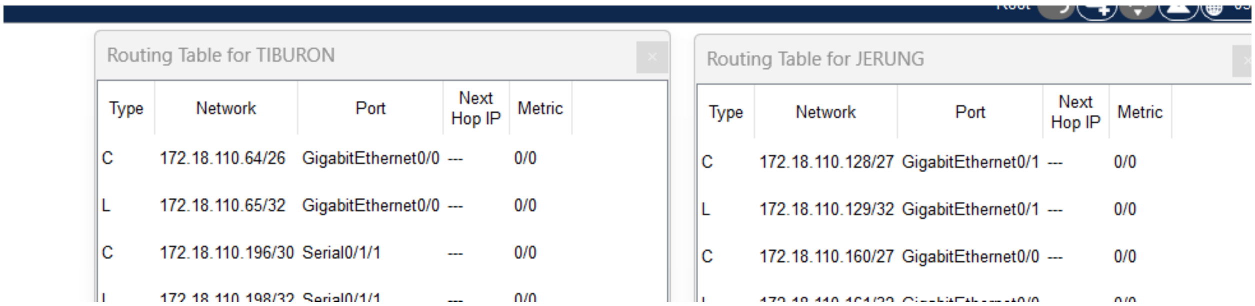


Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

   a. What can you say about the addresses used in the 'network' instructions in Figure E?
      172.18.110.128 connects to the TIBURON LAN subnet.
      172.18.110.160 is not an IP for any subnets.
      172.18.110.192 connects to the JERUNG-SHARK connection subnet.
      172.18.110.196 connects to the JERUNG-TIBURON connection subnet.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64

TIBURON (config-router) #network 172.18.110.196

TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

a. What are the changes seen in TIBURON?
There are new routes in the routing table.

b. What are the Networks with type R in TIBURON and JERUNG?
JERUNG

- 172.18.110.128/26
- 172.18.110.200/30

TIBURON

- 172.18.110.0/27
- 172.18.110.32/27
- 172.18.110/192/30

c. Ping PC3 from PC1. Was it successful?
Yes.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping172.18.110.190
Invalid Command.

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

d. Ping PC4 from PC1. Was it successful?
It was not successful.

```
C:\>ping 172.18.11.126

Pinging 172.18.11.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.11.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

e. Explain the reasons for your answer.
The network to SHARK LAN, where PC4 is located in, is not connected.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>en
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.32
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#
```

g. Open router SHARK's routing table and paste here.

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=15ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 15ms, Average = 4ms

<table>
<tr>
<td>2</td>
<td>
<pre>
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=2ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=2ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
</pre>
</td>
</tr>
</table>

3	C:\>ping 172.18.110.190
	
	Pinging 172.18.110.190 with 32 bytes of data:
	
	Reply from 172.18.110.190: bytes=32 time=18ms TTL=126
	Reply from 172.18.110.190: bytes=32 time=13ms TTL=126
	Reply from 172.18.110.190: bytes=32 time=2ms TTL=126
	Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
	
	Ping statistics for 172.18.110.190:
	    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
	Approximate round trip times in milli-seconds:
	    Minimum = 1ms, Maximum = 18ms, Average = 8ms
	
	C:\>ping 172.18.110.190
	
	Pinging 172.18.110.190 with 32 bytes of data:
	
	Reply from 172.18.110.190: bytes=32 time=16ms TTL=126
	Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
	Reply from 172.18.110.190: bytes=32 time=7ms TTL=126
	Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
	
	Ping statistics for 172.18.110.190:
	    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
	Approximate round trip times in milli-seconds:
	    Minimum = 1ms, Maximum = 16ms, Average = 6ms

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
    a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24
    b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

    c. After this change, can PC4 and PC1 ping PC3?
    No. Destination host is unreachable.

    d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

    e. Referring to the routing table, explain your findings.
    The router knows the way to the subnet 170.18.110.128 but not the way to the new network address.

    f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
    Advertise 192.168.1.0/24 network via RIP from TIBURON.

g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.0
TIBURON(config-router)#network 172.18.110.32
TIBURON(config-router)#network 172.18.110.64
```

h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

<table>
<tr><th>PC</th><th>Ping result</th></tr>
<tr><td>1</td><td><pre>C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=21ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 21ms, Average = 6ms</pre></td></tr>
<tr><td>2</td><td><pre>C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=28ms TTL=126
Reply from 172.18.110.62: bytes=32 time=2ms TTL=126
Reply from 172.18.110.62: bytes=32 time=14ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 28ms, Average = 11ms</pre></td></tr>
</table>

4

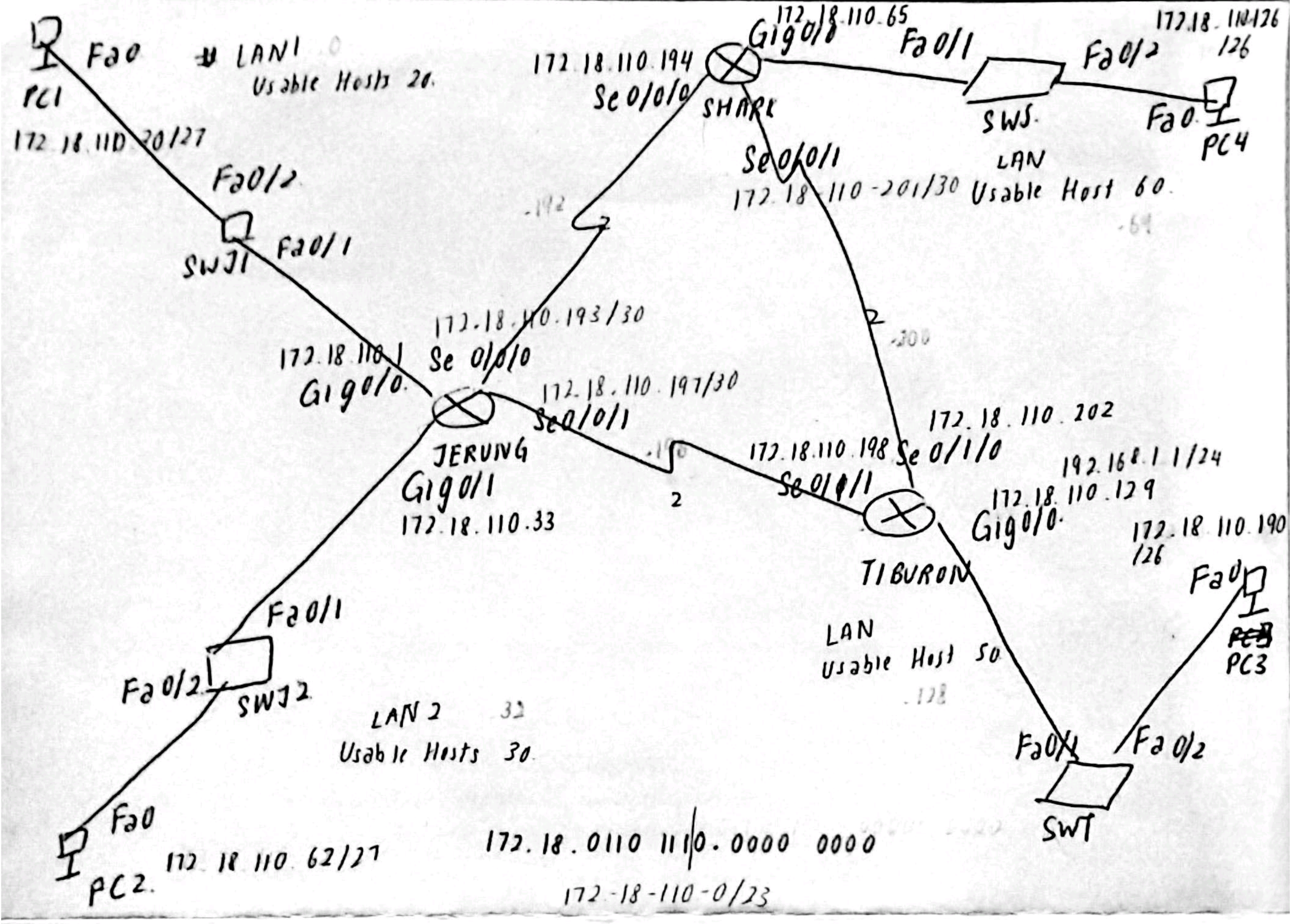
```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=18ms TTL=126
Reply from 172.18.110.126: bytes=32 time=4ms TTL=126
Reply from 172.18.110.126: bytes=32 time=10ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 18ms, Average = 8ms
```

REFLECTION

What have you learned in this task?

I have learned a lot about routing and configurating the routers and PC. I get to see how the network address and IPs work together and communicate between devices. Besides, I also realised the importance of correctly configurating interfaces. I have mistakenly swapped the IP of TIBURON Se0/1/0 with TIBURON Se0/1/0, which caused unsuccessful attempts to ping PC3 from other PCs after configuring the routing protocol. Other than that, I also learned that changing the IP of a network address require reconfiguration for all the devices within the subnet. Thus, subnetting is vital to reduce rework when we configure a network.

The following appended are the workings done throughout the lab.



JERUNG LAN1

2^n = IPs 12

Usable Hosts = 20

Subnet Size = $2^n \geq 20+2$

$2^5 \geq 22$

Subnet Mask = ~~120~~ /27

Network Address = 172.18.110.0/27

172.18.110.000|0 0000

172.18.110.0 - 172.18.110.31

JERUNG LAN 2

Usable Hosts = 30

Subnet size = $2^n \geq 30+2$

$2^5 \geq 32$

Subnet Mask = /27

Network Address = 172.18.110.32/27

172.18 110.001|0 0000

172.18.110.32 - 172.18.110.63

SHARK LAN

Usable Hosts = 60

Subnet size = $2^n \geq 60+2$

$2^6 \geq 62$

Subnet mask = /26

Network Address = 172.18.11.64

172.18.110.01|00 0000

172.18.11.64 - 172.18.11.127

TIBURON LAN

Usable Hosts = 50.

Subnet size = $2^n \geq 50+2$

$2^6 \geq 62$

Subnet mask = /26

Network Address = 172.18.110.128

172.18.110.10|00 0000

172.18.110.128 - 172.18.110.191

JERUNG - SHARK

Usable Host = 2

Subnet size = $2^n \geq 2+2$

$2^2 \geq 4$

Subnet mask = /30

Network Address = 172.18.11.192

172.18.110.<u>1100</u> 00|00

172.18.110.192 - 172.18.110.195

SHARK - TIBURON

Usable Host = 2

Subnet mask = /30

Network Address = 172.18.11.200

172.18.110 1100 <u>10</u>|00

172.18.11.200 - 172.18.11.203

JERUNG - TIBURON

Subnet mask = /30

Network Address = 172.18.11.193

172.18.110.1100 <u>01</u>|00

172.18.110.196 - 172.18.110.199.

JERUNG

172.18.110.128 → TIBURON LAN,

172.18.110.160 →

172.18.110.192 → JERUNG - SHARK

172.18.110.196 → JERUNG - TIBURON,

SHARK

172.18.110.0 → JERUNG LAN1

172.18.110.32 → JERUNG LAN2

172.18.110.192 → JERUNG SHARK

172.18.110.200 → TIBURON - SHARK

TIBURON

172.18.110.64 → SHARK LAN

172.18.110.196 → JERUNG - TIBURON,

172.18.110.200 → TIBURON - SHARK

NetAd 192.168.1.0/24 192.168.1.|0000 0000

Gig 0/0 192.168.1.1/24

192.168.1.0 - 192.168.1.255

PC3 : 192.168.1.254